ISWE406P – Agile Development Process and Devops

Lab Assignment IV

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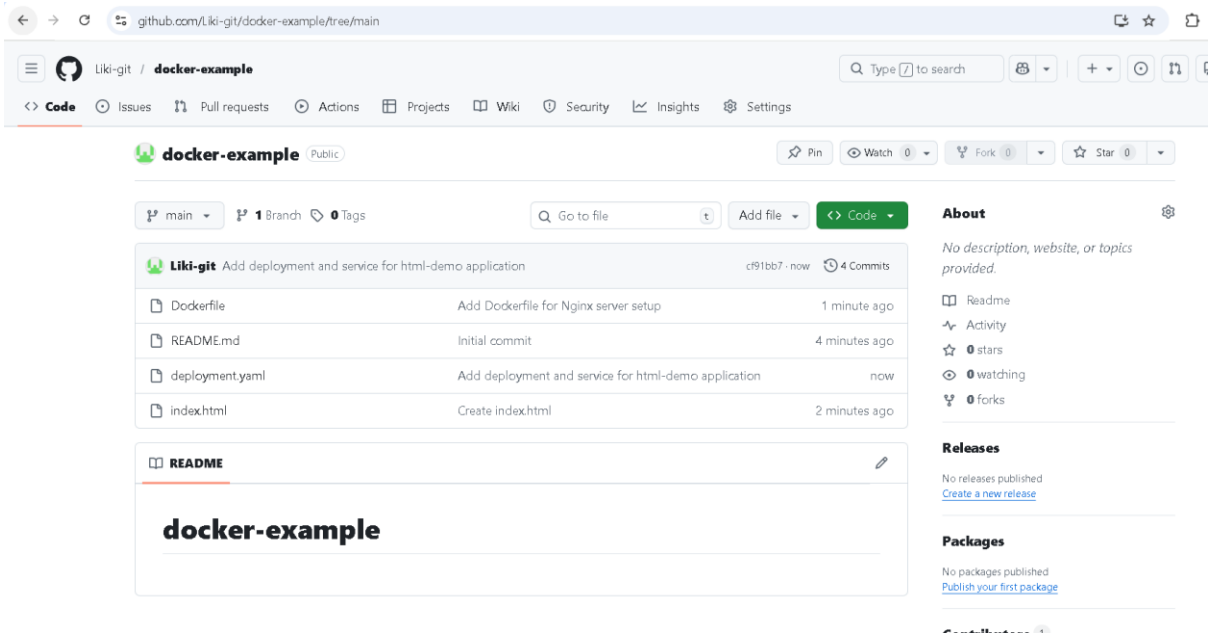
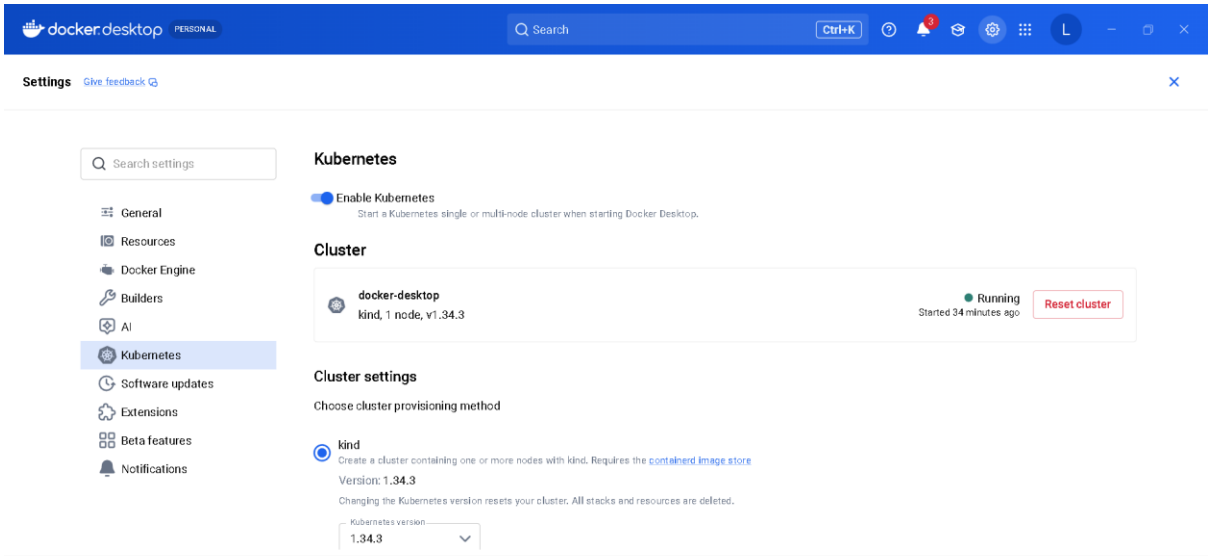
SLOT : L49+L50

1. Create a repository in GIT and create the following files:

- index.html
- Dockerfile
- Deployment.yaml

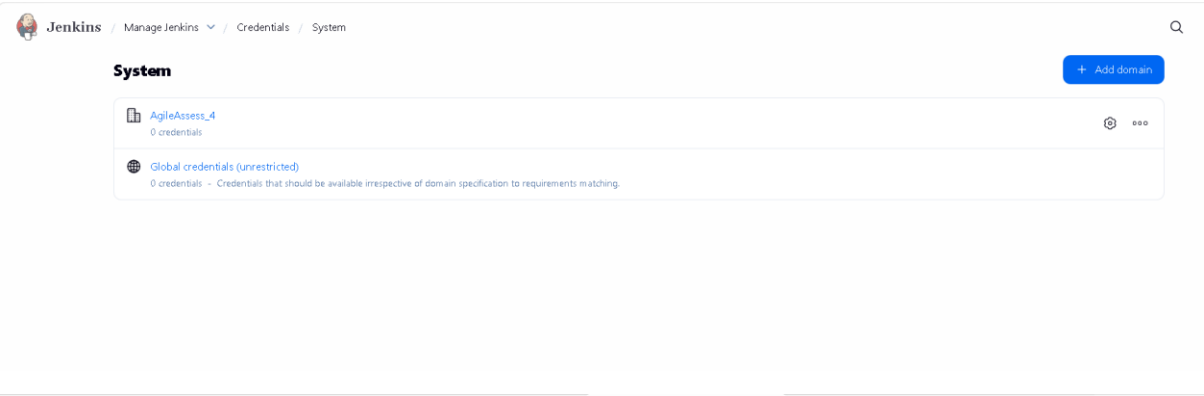


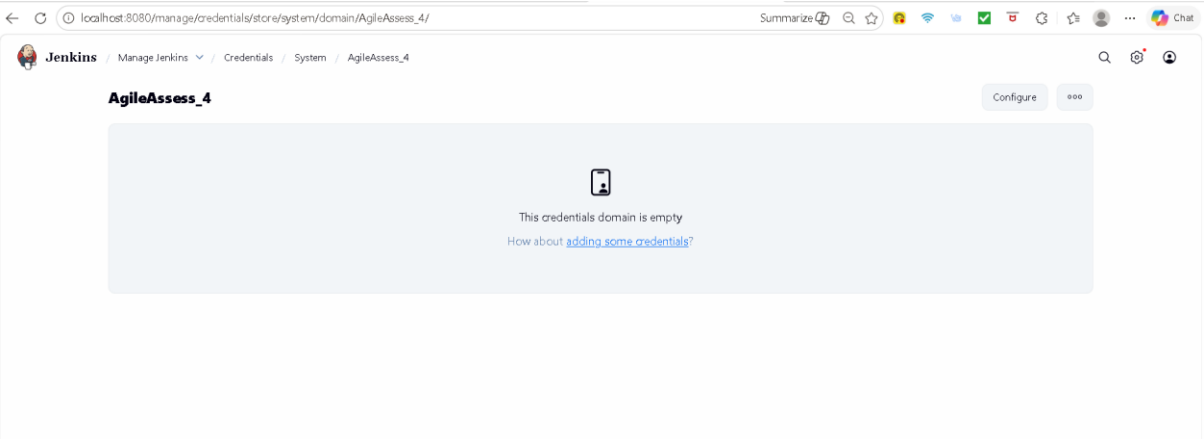
2. Enable Kubernetes



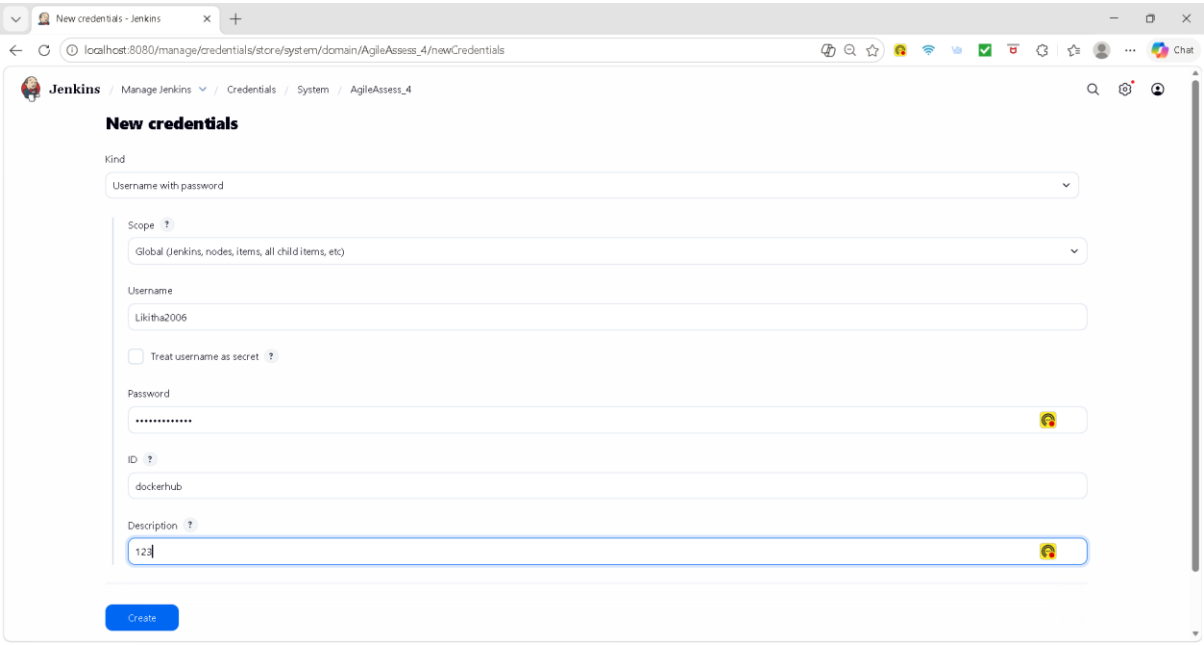
3. Open Jenkins

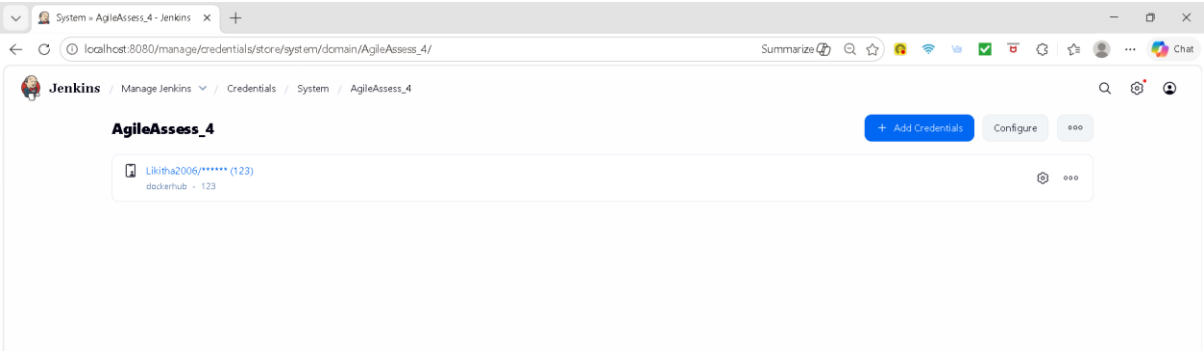
In Jenkins Setting --> Credentials



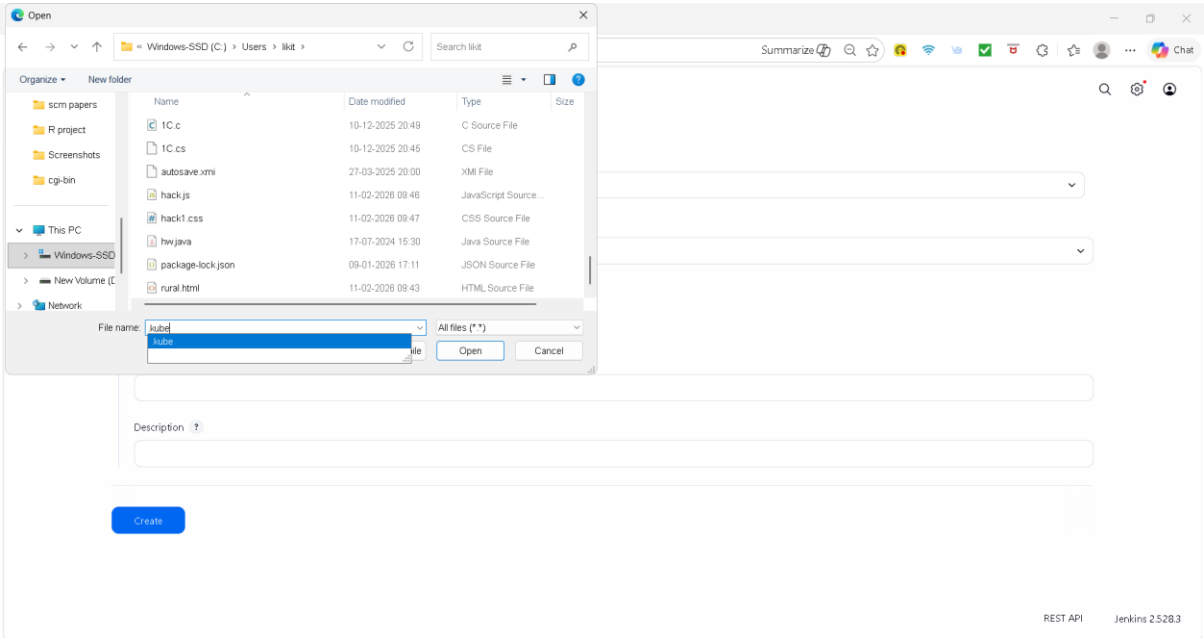


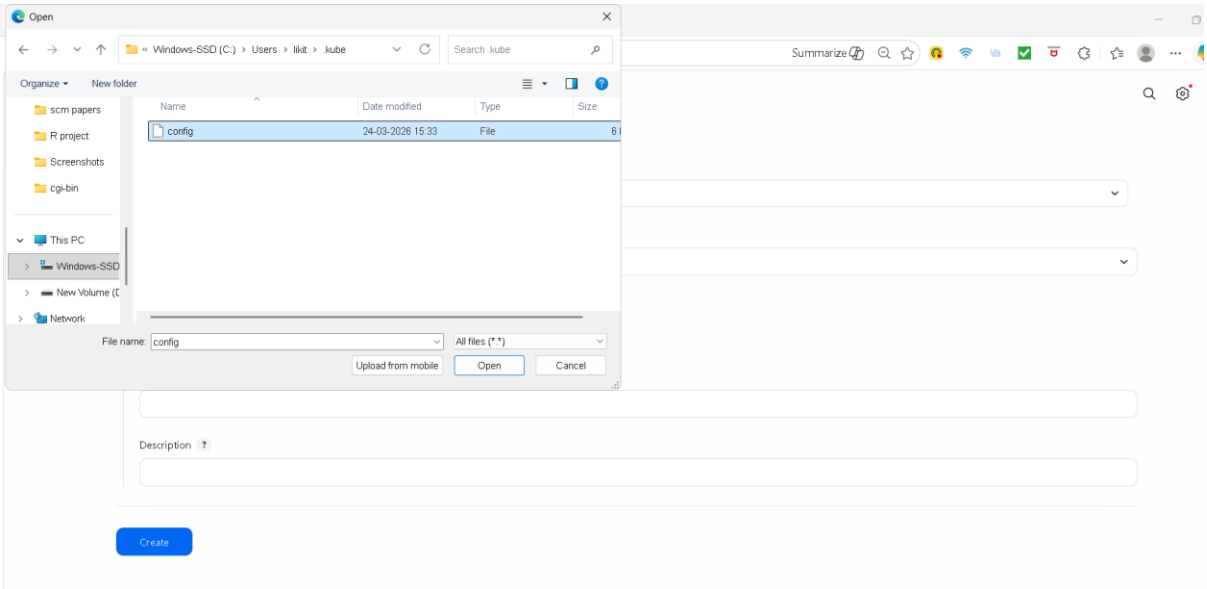
1st Credential

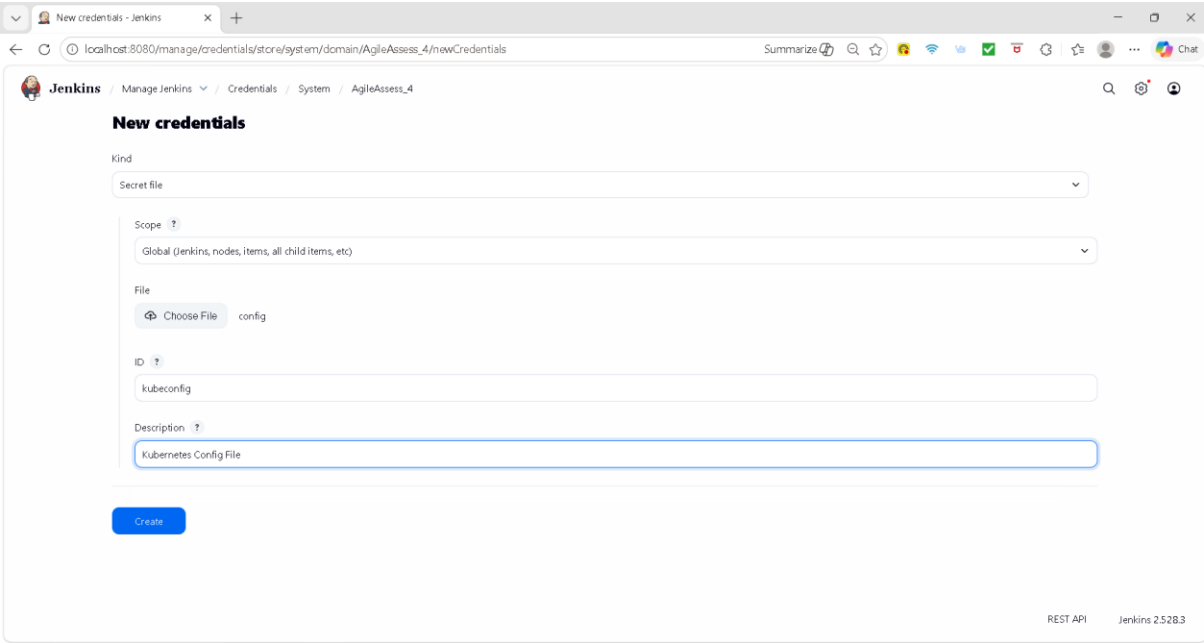


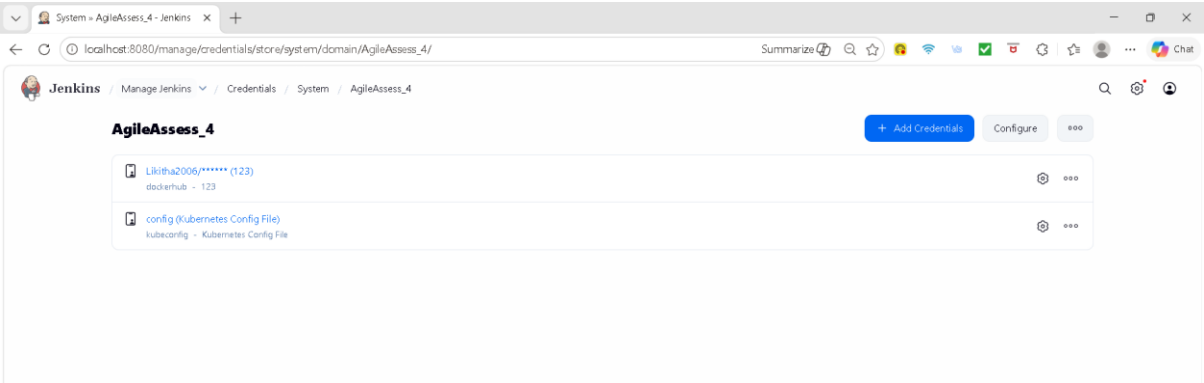


2nd Credential (Secret file)-.kube file

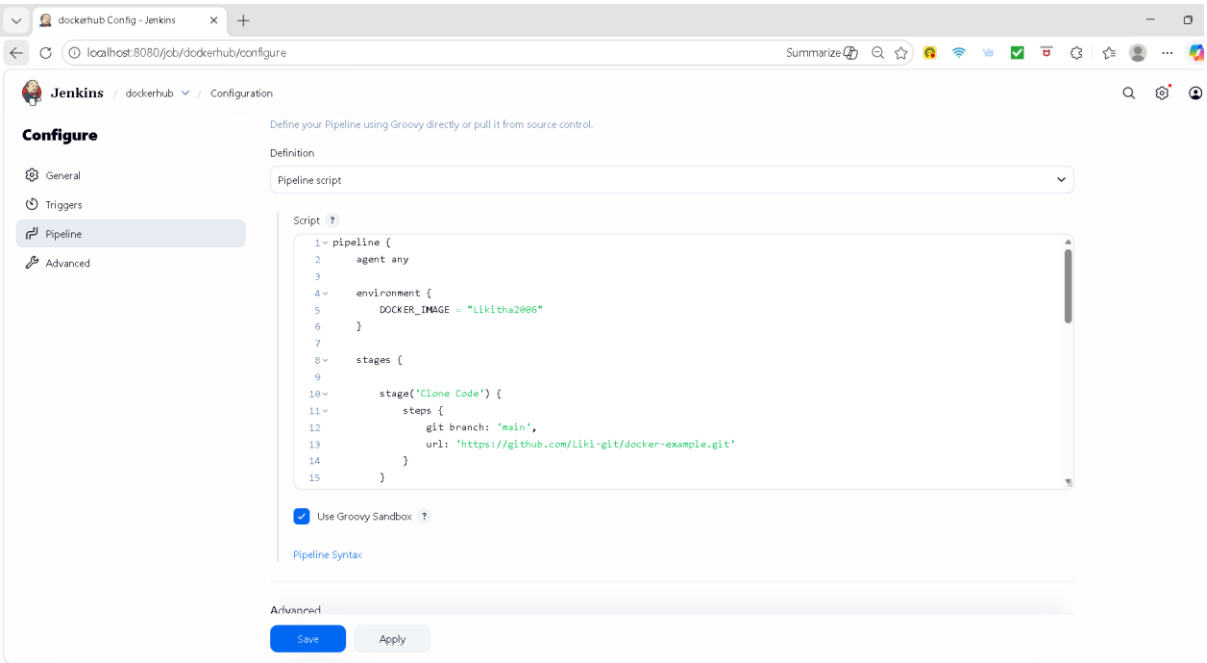








Create New Item -> Pipeline , pipeline script



Build successful

dockerhub - Jenkins

localhost:8080/job/dockerhub/

Jenkins / dockerhub

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Stages
- Rename
- Pipeline Syntax

Builds

Filter

Today

#6 2:18 PM

#5 2:09 PM

dockerhub

Add description

Stage View

	Clone Code	Build Docker Image	Push Image	Deploy to Kubernetes
Average stage times: (full run time: ~4min 25s)	3s	6s	2min 6s	1s
#6 Mar 26 14:18 No Changes	3s	6s	4min 11s	2s
#5 Mar 26 14:09 No Changes	3s	6s	2s failed	154ms failed

Permalinks

- Last build (#6), 10 min ago
- Last stable build (#6), 10 min ago
- Last successful build (#6), 10 min ago
- Last failed build (#5), 20 min ago
- Last unsuccessful build (#5), 20 min ago
- Last completed build (#6), 10 min ago

Jenkins / dockerhub / #6 / Console Output

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#6'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

Console Output

Download | Copy | View as plain text

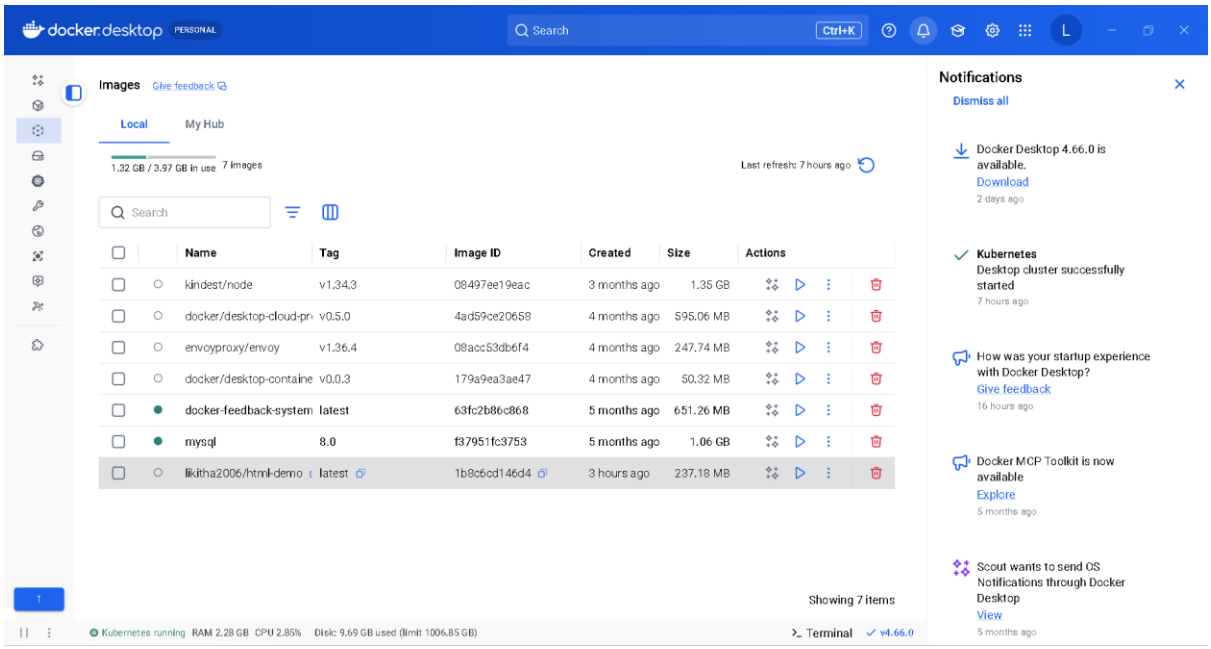
```
Started by user P Likitha
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\likit\.jenkins\workspace\dockerhub
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > C:\Program Files\Git\bin\git.exe rev-parse --resolve-git-dir C:\Users\likit\.jenkins\workspace\dockerhub\.git # timeout=10
Fetching changes from the remote Git repository
 > C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/Liki-git/docker-example.git # timeout=10
Fetching upstream changes from https://github.com/Liki-git/docker-example.git
 > C:\Program Files\Git\bin\git.exe --version # timeout=10
 > git --version # 'git version 2.53.0.windows.2'
 > C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/Liki-git/docker-example.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
 > C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 3c024bdb64785d6e794b899520bc53b0ff108d23 (refs/remotes/origin/main)
 > C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
 > C:\Program Files\Git\bin\git.exe checkout -f 3c024bdb64785d6e794b899520bc53b0ff108d23 # timeout=10
 > C:\Program Files\Git\bin\git.exe branch -a -v --no-abbrev # timeout=10
 > C:\Program Files\Git\bin\git.exe branch -D main # timeout=10
 > C:\Program Files\Git\bin\git.exe checkout -b main 3c024bdb64785d6e794b899520bc53b0ff108d23 # timeout=10
Commit message: "Update deployment.yaml"
 > C:\Program Files\Git\bin\git.exe rev-list --no-walk 3c024bdb64785d6e794b899520bc53b0ff108d23 # timeout=10
```

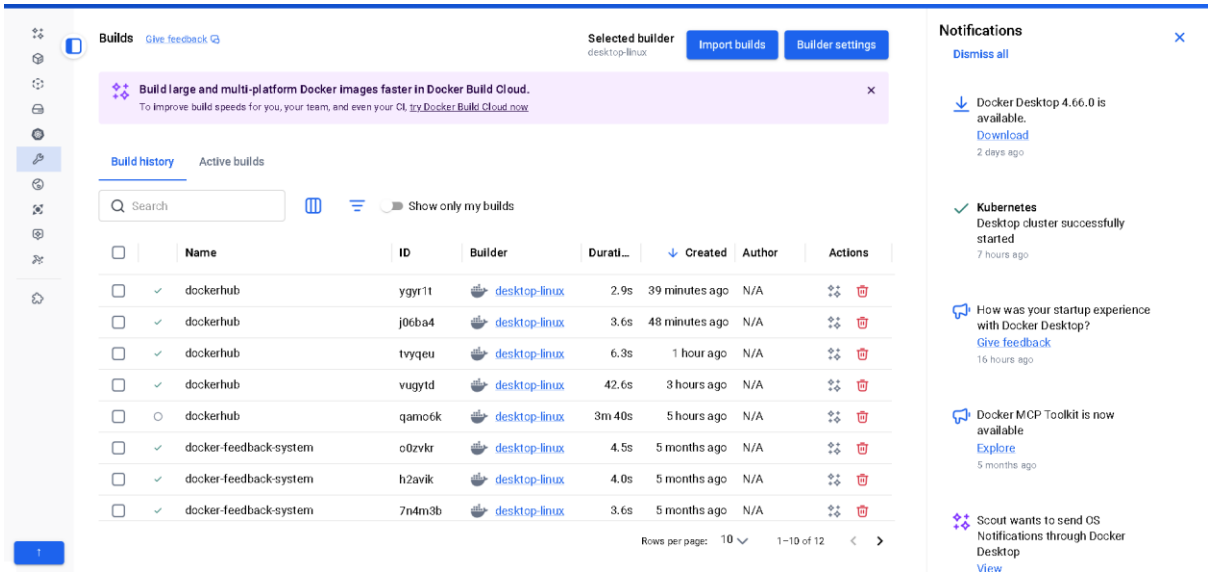
Jenkins / dockerhub / #6 / Console Output

```
3a02375f860f: Pushed
587e3d84dbb5: Pushed
cde7a05ae428: Pushed
bb3d0aa29654: Pushed
ec781dee3f47: Pushed
latest: digest: sha256:1b8c6cd146d42b811cc20906447071cc8243eccc346c64dda955890979f3ff0d size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] withCredentials
Masking supported pattern matches of %KUBECONFIG%
[Pipeline] {
[Pipeline] bat

C:\Users\likit\.jenkins\workspace\dockerhub>set KUBECONFIG=****

C:\Users\likit\.jenkins\workspace\dockerhub>kubectl apply -f deployment.yaml --validate=false
deployment.apps/html-demo created
service/html-demo-service created
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

```
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\docker-example> git push
Everything up-to-date
PS                    neDrive\Desktop\Agile_asses1\docker-example>
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\docker-example>
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\docker-example> kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
html-demo-6fd9667bc6-kkqps    1/1     Running   0          17m
html-demo-6fd9667bc6-rrlgn    1/1     Running   0          17m
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\docker-example> kubectl get services
NAME                 TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
html-demo-service    NodePort    10.96.194.97   <none>        80:30010/TCP   17m
kubernetes           ClusterIP   10.96.0.1      <none>        443/TCP        47h
```

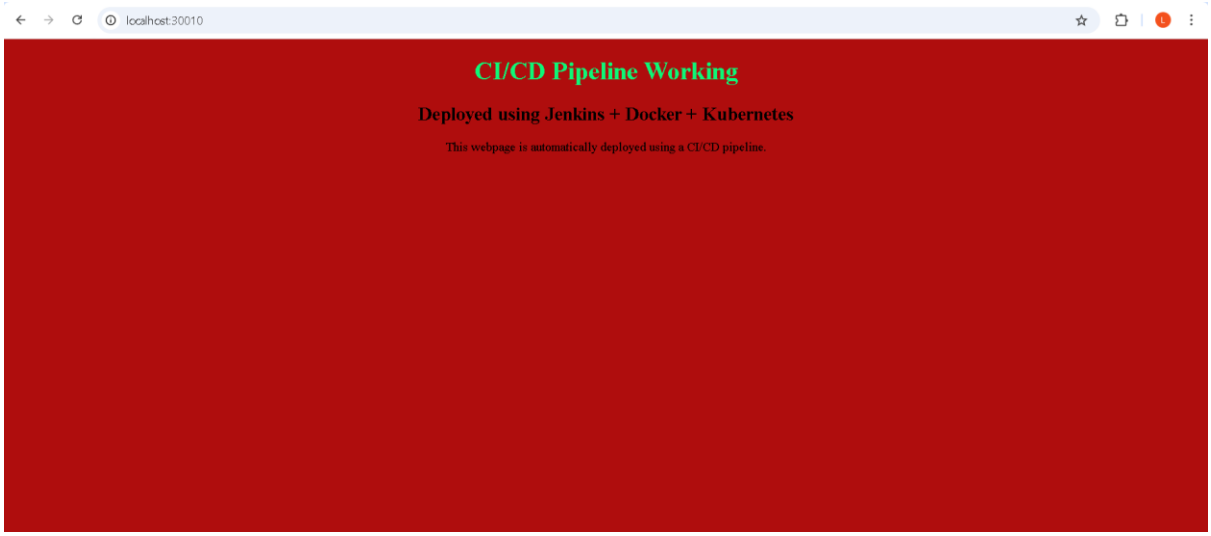
Docker desktop view





Finally,

HTML page should open in the port mentioned in deployment.yaml file: nodePort: 30010


localhost:30010
CI/CD Pipeline Working
Deployed using Jenkins + Docker + Kubernetes
This webpage is automatically deployed using a CI/CD pipeline.